



Quality of life of children with short bowel syndrome and their caregivers from the perspective of caregivers: a cross-sectional study in China

Haixia Feng¹ · Qingqing Wu¹ · Chen Cai¹ · Weihui Yan^{1,3} · Shanshan Chen¹ · Qinzhou Xu¹ · Shiyi Liu¹ · Wei Cai^{2,3} · Guannan Bai⁴ · Ying Wang^{1,3}

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Abstract

Short bowel syndrome (SBS) is the most common cause of intestinal failure, requiring long-term of parenteral nutrition support. The objectives of this study were to evaluate health-related quality of life (HRQoL) of children with SBS and their caregivers, as well as the association with burden and depression of caregivers. Data were collected through a cross-sectional study using the proxy-reported PedsQL 4.0 Generic Core Module (PedsQL GCM) and PedsQL 3.0 Family Impact Module (FIM) to assess the HRQoL of children and their caregivers. Differences in HRQoL scores across subgroups were assessed using two-sample *t*-tests, one-way ANOVA. The Zarit Burden Interview (ZBI) and the Patient Health Questionnaire (PHQ-9) were employed to assess the burden and depression of the caregivers. Ninety-six children with SBS and their caregivers participated in the study. Compared with established norms for healthy children, children with SBS aged 1 to 24 months exhibited significantly lower scores in the physical health summary ($p < 0.01$). Children with SBS aged 2 to 18 years scored an average of 5.99 points lower in the psychosocial health summary ($p = 0.01$) and 4.56 points lower in the total score ($p = 0.039$). Families of a child with SBS exhibited significantly lower scores across nearly all dimensions of the PedsQL 3.0 FIM compared with those of healthy controls ($p < 0.05$). In multivariate logistic regression analysis, younger age, male gender, long duration of hospitalization, perceived financial burden were independently associated with a decreased total score on the PedsQL FIM ($p < 0.05$). The Spearman correlation analysis revealed that the PHQ-9 and the ZBI total score were negatively correlated with PedsQL 4.0 GCM scale score as well as total scale ($p < 0.02$) among children aged 2–18 years. **Conclusion:** This study provides a comprehensive analysis of HRQoL among children with SBS and their caregivers and identifies multiple associated predictors. The burden of caregivers and depression affect the HRQoL of children with SBS. We highlight the importance of providing increased attention and support not only to the patients but also to their caregivers.

What is Known:

- Pediatric-onset short bowel syndrome (SBS) is linked to reduced health-related quality of life (HRQoL) of both patients and their families. The influencing factors and the correlation between HRQoL and the levels of depression and caregivers burden remains inadequately explored.

What is New:

- The independent predictors of a higher total score on the PedsQL FIM were older age, female gender, short duration of hospitalization and perceived no financial burden. The caregiver's high burden and depression showed a negative correlation with the children's HRQoL.

Keywords Health-related quality of life · Short bowel syndrome, · Burden · Depression

Introduction

Short bowel syndrome (SBS) is the most common cause of intestinal failure (IF) [1]. Although the application of PN has reduced the mortality rate of children with SBS [2], long-term PN is associated with an increased risk of complications. Children with SBS often require long-term

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Haixia Feng and Qingqing Wu contributed equally to this work.

Extended author information available on the last page of the article

hospitalization. This form of management places substantial demands on caregivers in terms of time and energy, significantly disrupting their daily lives, including professional and social activities.

Over the past few decades, substantial advancements have been achieved in the treatment of SBS. Nevertheless, many clinicians remain skeptical about the long-term health-related quality of life (HRQoL) outcomes for children with SBS [3]. By regularly assessing quality of life, researchers can monitor treatment outcomes, make timely adjustments to treatment plans, and gain insight into the needs of children and their caregivers, thereby enhancing the overall HRQoL in pediatric populations. To date, studies conducted abroad have investigated the HRQoL of children with SBS and their caregivers [4–8]. These studies demonstrated that the overall HRQoL of children with SBS was significantly lower compared to that of healthy peers and even lower than that of children with other chronic diseases, particularly in the dimensions of physical and mental health, as well as social and school functioning. Parents and children often reported a decline in HRQoL [6, 8–10]. Notably, a recently published study on the long-term trajectory of HRQoL in children with IF revealed no significant changes over time [11]. Furthermore, the study indicated that although the condition of the children has a significant impact on the lives of families, many families have established a systematic daily care routine to effectively integrate child care into daily life, thereby achieving a balanced coexistence of family life and caregiving [12]. The heterogeneity observed in previously published articles may be attributed to relatively small sample sizes and variations in public healthcare systems across countries. Families of children with SBS are also profoundly affected by the condition; however, HRQoL in families of children with IF has not been sufficiently investigated. Moreover, the correlation between HRQoL and the levels of depression and caregivers burden remains inadequately explored.

In China, data regarding the HRQoL of children with SBS and their caregivers remain limited. The primary objective was to assess the HRQoL of children with SBS and their caregivers and to identify the associated predictors. The secondary objective was to examine the depression and burden among caregivers and to explore their potential associations with the HRQoL of children.

Methods

Research design and participants

This was a cross-sectional study involving SBS children who were treated at the Division of Pediatric Gastroenterology and Nutrition, Xinhua Hospital Affiliated to Shanghai Jiao Tong University School of Medicine between January

2007 and June 2024. The inclusion criteria were as follows: (1) age under 18 years old and diagnosed of SBS; (2) consent was provided by the legal guardian. Exclusion criteria included unstable vital signs, presence of catheter-related bloodstream infection within the past month, or any condition that might interfere with the study's objectives.

Demographic and clinical characteristics

Basic information was collected, including age, gender, employment status, family income, insurance coverage, and SBS treatment costs. Clinical data were also recorded, including the etiology, surgical history, duration of hospitalization, number of catheter-related infections, and the types of nutritional support.

Health-related quality of life

The PedsQL™ Infant Scales [13] were designed as a generic HRQoL instrument tailored specifically for both healthy and ill infants aged 1–24 months. The 36-item (ages 1–12 months) and 45-item (ages 13–24 months) PedsQL™ Infant Scales (Physical Functioning, Physical Symptoms, Emotional Functioning, Social Functioning, Cognitive Functioning) were completed by the caregivers. HRQoL data in the SBS children aged 2–18 years old were collected using the proxy-reported PedsQL 4.0 Generic Core Module (PedsQL GCM) [14], designed to measure dimensions of health: physical-, emotional-, social-, and school functioning, in addition to two multidimensional scores of psychosocial health and total HRQoL. The PedsQL 3.0 Family Impact Module (FIM) was employed to evaluate the impact of SBS on the caregiver's QoL and family functioning over the past month. Physical, emotional, social, and cognitive functioning; communication; worry; daily activities, and family relationships were measured. Both instruments have been validated in China with good reliability and validity [15, 16] and completed by caregivers. Each score was transformed into a 0–100 scale, so that higher PedsQL scores indicated higher HRQoL.

Burden and depression among caregivers

The Zarit Burden Interview (ZBI) [17] is a widely recognized 22-item questionnaire designed to assess caregivers' subjective burden. The Chinese version demonstrated satisfactory reliability and validity in a previous study [18]. Caregiver burden scores were classified into four categories: trivial or no burden (0–20), mild burden (21–30), moderate burden (31–40), and severe burden (41–88). The Patient Health Questionnaire (PHQ-9) is a 9-item instrument based on symptoms of major

depressive disorder symptoms, designed to screen and evaluate the severity of depression in primary care and other medical settings. A total score of 10 or higher indicates major depression, while a score of 5 or higher suggests mild depression [19].

Statistical analysis

All statistical analyses were performed using SPSS Statistics version 22 (IBM Corp., NY, USA). For descriptive analysis, continuous variables were summarized using means and standard deviations (SD) or medians and interquartile ranges (IQR) depending on the data distribution, while categorical variables were described in terms of frequencies and proportions. One-sample *t*-tests were conducted to compare the average scores of various scales in our cohort with previously published normative data from healthy controls [13, 16, 20]. We employed two-sample *t*-tests and one-way ANOVA to assess mean differences in PedsQL scores across groups, based on patients' demographic and clinical characteristics. Univariate analysis was used to assess the significance of each variable. Subsequently, all variables that demonstrated significance in the univariate analysis were included in a multivariate logistic regression model to identify the independent predictors of HRQoL. Odds ratios (ORs) and 95% confidence intervals (CI) were calculated. Lastly, Spearman correlation analysis was utilized to assess the associations between children's HRQoL and both caregiver burden and depression scales. A two-sided test *p*-value less than 0.05 was considered statistically significant. We also evaluated the clinical relevance using Cohen's effect size (Cohen's *d*).

Results

Characteristics of the study population

One hundred and twenty patients were enrolled, of whom 112 patients were eligible and assessable. In all, a total of 96 caregivers of children with SBS who completed the questionnaire participated in this study. The demographic characteristics of the participants are summarized in Table 1. The average age of the children was 62.11 ± 44.70 months. 59.4% of families reported spending more than €24,400 on SBS-related treatment, which refers to out-of-pocket medical costs. The average remaining small bowel length was 53.31 ± 22.63 cm. Twenty-six children (27.1%) were dependent on PN. 8 patients (8.3%) received teduglutide therapy and 5 patients (5.2%) underwent serial transverse enteroplasty (STEP) procedure. Forty-six patients (47.9%) required more than 4 hospital admissions and 58 patients (60.4%) had a total hospitalization period exceeding 6 months (see Table 2).

HRQoL scores and comparison to reference populations

Compared with existing norms for healthy children [13, 16], subjects aged 1 to 12 months scored significantly lower on the physical health summary (73.18 ± 12.49 vs 84.98 ± 9.45 , $p < 0.01$, effect size 0.94). Similarly, children aged 13–24 months also scored significantly lower than healthy children on the physical health summary ($p < 0.01$, effect size 1.40) and total score ($p = 0.005$). Compared with reference values from healthy children, study subjects with SBS aged 2 to 18 years scored significantly lower in multiple domains of HRQoL: social functioning was reduced by an average of 8.83 points ($p = 0.003$), school functioning by 9.93 points ($p < 0.01$), psychosocial health summary by 5.99 points ($p = 0.01$), and total score by 4.56 points ($p = 0.039$) (see Table 3).

Family Impact Module

We compared the mean values of children aged 5–12 years in our study with those reported for children of the same age group in the existing literature [20]. Families of children with SBS exhibited significantly lower scores across nearly all dimensions of the PedsQL 3.0 FIM compared to families of healthy children ($p < 0.05$). The mean (SD) total score for the PedsQL 3.0 FIM was 70.26 (20.68), which was significantly lower than the score of 80.07 (13.98) observed in families with healthy children (shown in Table 4).

Differences in mean scale/summary scores and the average total score of PedsQL 4.0 GCM and 3.0 FIM

eTable 1 presents the differences in mean scale and the total score of the PedsQL 4.0 GCM among children with SBS. Children with caregivers not perceiving a financial burden exhibited significantly higher mean scores in physical functioning ($p < 0.01$), social functioning ($p < 0.01$), school functioning ($p = 0.04$), and total scale ($p < 0.001$) of the PedsQL 4.0 GCM. Children without ileocecal valve and those with PN duration exceeding 1 year showed relatively lower mean school functioning scores ($p = 0.003$ and $p = 0.002$, respectively). Additionally, children with more than 4 hospitalizations had lower scores in emotional function and total score than those with fewer than 4 hospitalizations ($p = 0.007$ and $p = 0.023$, respectively). The total score was significantly lower in patients receiving teduglutide treatment than in those not receiving the intervention ($p = 0.044$). Compared with children having fewer than 2 daily bowel movements, those with more than 1 daily bowel movements demonstrated significantly lower physical functioning scores

Table 1 General characteristics of children with SBS ($N=96$)

Variables	Values
Age, months, mean $\pm$ SD	62.11 $\pm$ 44.70
Age group, n (%)	
0–12 months	11 (11.5)
1–2 years	15 (15.6)
2–4 years	21 (21.9)
5–7 years	25 (26.0)
8–12 years	19 (19.8)
13–18 years	5 (5.2)
Female, n (%)	35 (36.5)
The caregiver who filled in the questionnaire, n (%)	
Mother	61 (63.5)
Father	32 (33.3)
Others	3 (3.1)
Employment status, n (%)	
Both are employed	55 (57.3)
Only father is employed	34 (35.4)
Only mother is employed	4 (4.2)
Unemployed	3 (3.1)
Having someone to take care of the child except parents, yes, n (%)	64 (66.7)
Annual household income, EUR, n (%)	
< 12,200	45 (46.9)
12,200–24,4000	26 (27.1)
24,4000–61,000	21 (21.9)
> 61,000	4 (4.2)
Amount of out-of-pocket expenses for treating SBS	
< 6100	0 (0)
6100–12,200	10 (10.4)
12,200–24,4000	29 (30.2)
> 24,4000	57 (59.4)
Perceived financial burden due to SBS and treatments, yes, n (%)	83 (86.5)
Having medical insurance, yes, n (%)	94 (97.9)
Receiving social benefits/bonuses/subsidies, yes, n (%)	35 (36.5)
Schooling status, n (%)	
In school	45 (46.9)
Preschooler	51 (53.1)

SBS short bowel syndrome, SD standard deviation

($p=0.009$), social functioning scores ($p=0.016$) and total scores ($p=0.046$).

eTable 2 presents differences in the mean scale score of PedsQL 3.0 FIM and total scores across certain groups. Univariate analysis revealed that caregivers of male gender, infant subgroups, receipt social benefits, perceived financial burden, PN dependent patients reported lower scores in total score ($p<0.05$). Notably, caregivers of children experiencing recurrent catheter-related infections, prolonged hospital stays, or frequent hospitalizations also showed a significant reduction in mean scores across most dimensions and the total scales ($p<0.05$).

Multivariable regression data

Employing multivariate linear regression models, caregivers who perceived no financial burden were independently associated with a significant higher total score on the PedsQL 4.0 GCM ($p=0.022$). Moreover, PN duration increased by per month, resulting in a total score decrease of 0.12 points ($p=0.011$).

As is shown in eTable 3, the associations between selected variables and the total score of the PedsQL 3.0 FIM are based on multivariate linear regression models. Compared with boys, girls had a statistically significantly higher total score on the PedsQL 3.0 FIM ($\beta=8.46$; 95% CI=0.39,

Table 2 Clinical characteristics of children with SBS ($N=96$)

Variables	Results
Birth weight, kg, mean $\pm$ SD	2.55 $\pm$ 0.87
Gestational age, weeks, mean $\pm$ SD	35.55 $\pm$ 3.68
Premature, yes, n (%)	47 (49.0)
Etiology, n (%)	
Intestinal atresia	26 (27.1)
Volvulus and malrotation	26 (27.1)
NEC	27 (28.1)
Other	17 (17.7)
Remaining intestine, cm, median (interquartile range)	53.31 $\pm$ 22.63
No. abdominal operations	
≤ 3	22 (22.9)
> 4	74 (77.1)
Neonatal onset of illness, yes, n (%)	72 (75)
Length of PN therapy, months, median (IQR)	8.0 (5.0, 13.75)
Wean off PN, n (%)	70 (72.9)
Other structural malformations, yes, n (%)	14 (14.6)
Nutritional status (BMI z-score)	
z-score ≥ -1	51 (53.1)
$-2 \leq$ z-score < -1	21 (21.9)
$-3 \leq$ z-score < -2	8 (8.3)
z-score < -3	16 (16.7)
Teduglutide treatment	8 (8.3)
STEP	5 (5.2)
Number of catheter-related infections	
< 2	43 (44.8)
≥ 2	53 (55.2)
Number of hospitalizations	
≤ 4	50 (52.1)
> 4	46 (47.9)
Average length of hospital stay, months	
≤ 6	38 (39.6)
> 6	58 (60.4)

PN parenteral nutrition, BMI body mass index, NEC necrotising enterocolitis, IQR interquartile range, STEP serial transverse enteroplasty

16.52; $p=0.04$). We also observed that age increases by per month, resulting in a total score increase of 0.103 points ($p=0.021$). Furthermore, caregivers who perceived no financial burden were independently associated with a significantly higher PedsQL 3.0 FIM total score ($p=0.012$). Moreover, the duration of hospitalization increased by per month, resulting in a total score decrease of 0.52 points ($p=0.014$).

Correlation analysis

PedsQL™ Infant Scale scores were significantly lower for depressed caregivers, particularly in the physical health summary (-18.2 , $p=0.01$) and emotional functioning scale (-19.31 , $p=0.026$). Meanwhile, the correlation analysis revealed that total score of PHQ-9 was negatively correlated with both the physical health summary and the emotional functioning scale ($r=-0.419$ and $r=-0.422$, respectively). PedsQL 4.0 GCM scale scores were lower when caregivers were depressed or experienced severe burden, across all scales and the total score, except for the social functioning scale for children aged 2–18 years. The PHQ-9 and the ZBI total score were negatively correlated with the proxy-reported PedsQL 4.0 GCM scale scores and the total scale ($p<0.02$) (Table 5 and eTable 4).

Discussion

HRQoL has emerged as a measurable health outcome in pediatric SBS, providing valuable guidance to clinicians and caregivers in making informed patient management decisions.

HRQoL of children with SBS and caregivers

The results of our cross-sectional study demonstrated that children with SBS exhibit significantly lower HRQoL scores compared to healthy subjects. These findings are consistent with those reported in several previous studies [4, 5, 8, 21, 22]. Interestingly, our results indicate that the decrease in HRQoL of infants is primarily attributable to the physical health summary component. However, for children aged 2–18 years, the difference is attributed to the lower scores in the psychosocial health summary. Infants are vulnerable to the adverse effects of digestive issues and malnutrition, both of which can hinder their developmental progress. For school-aged children, social interactions and learning processes assume greater significance. Given the significant differences observed in the domains of social functioning and school functioning, there is an opportunity within intestinal rehabilitation programs to develop targeted interventions aimed at enhancing these specific dimensions, thereby improving HRQoL [22]. Many problems relate to the presence of the venous access device and concern infections or vascular occlusion due to thrombosis [23]. In addition, the need for high-level care and frequent hospitalizations have disrupted daily activities [24]. The result also revealed that specific clinical characteristics, including the absence of ileocecal retention and a prolonged duration of PN therapy, were associated with lower scores in the school

Table 3 PedsQL infant scales or 4.0 Generic Core Scales compared with reference data for healthy children*

Characteristics	α	Children with SBS (mean $\pm$ SD)	Healthy children (mean $\pm$ SD)	<i>t</i> value	<i>P</i> value	Effect size (<i>d</i>)
		Infants with SBS (1–12 months, <i>N</i> = 11)	<i>N</i> = 246			
Physical health summary	0.846	73.18 $\pm$ 12.49	84.98 $\pm$ 9.45	− 3.99	< 0.01	0.94
Emotional functioning	0.939	72.16 $\pm$ 15.68	76.59 $\pm$ 13.71	− 1.04	0.298	0.28
Social functioning	0.711	84.65 $\pm$ 15.40	89.62 $\pm$ 14.87	− 1.08	0.280	0.32
Cognitive functioning	0.959	76.13 $\pm$ 22.50	83.11 $\pm$ 20.65	− 1.09	0.276	0.31
Psychosocial health summary	0.947	77.65 $\pm$ 15.89	80.47 $\pm$ 12.67	− 0.71	0.48	0.18
Total score	0.949	75.86 $\pm$ 13.92	82.47 $\pm$ 9.95	− 1.56	0.15	0.47
		Toddlers with SBS (13–24 months, <i>N</i> = 15)	<i>N</i> = 141			
Physical health summary	0.936	64.56 $\pm$ 17.3	88.84 $\pm$ 7.68	− 5.38	< 0.01	1.4
Emotional functioning	0.966	66.11 $\pm$ 21.31	78.60 $\pm$ 12.80	− 2.13	0.050	0.59
Social functioning	0.896	85.60 $\pm$ 19.70	91.14 $\pm$ 10.77	− 1.07	0.301	0.28
Cognitive functioning	0.960	72.70 $\pm$ 26.33	84.65 $\pm$ 15.76	− 1.73	0.105	0.45
Psychosocial health summary	0.959	74.85 $\pm$ 18.59	83.12 $\pm$ 11.02	− 1.69	0.111	0.44
Total score	0.973	70.74 $\pm$ 17.36	85.55 $\pm$ 8.74	− 3.26	0.005	0.85
		Children with SBS (2–18 years, <i>N</i> = 70)	<i>N</i> = 284			
Physical functioning	0.867	81.92 $\pm$ 18.27	82.18 $\pm$ 17.6	− 0.11	0.913	0.01
Emotional functioning	0.866	76.93 $\pm$ 17.24	76.90 $\pm$ 16.6	− 0.013	0.990	0.002
Social functioning	0.901	76.50 $\pm$ 22.67	85.33 $\pm$ 15.4	− 3.09	0.003	0.39
School functioning	0.693	68.10 $\pm$ 18.51	78.03 $\pm$ 16.5	− 3.88	< 0.01	0.53
Psychosocial health summary	0.895	74.10 $\pm$ 15.56	80.09 $\pm$ 17.6	− 2.61	0.01	0.34
Total score	0.913	76.18 $\pm$ 14.75	80.74 $\pm$ 16.9	− 2.07	0.039	0.27

SD standard deviation, α Cronbach's coefficient alpha

*In PedsQL 4.0 GCM for patients aged 2–18 years old, school functioning questions were optional and answered only when patients attending to school (*N* = 51)

The effect sizes for differences in means are categorized as small (0.20–0.49), medium (0.50–0.79), and large (0.80 and above)

Table 4 Comparing means of scale scores and total scores of PedsQL 3.0 FIM between patients with SBS and the healthy controls*

Characteristics	Children with SBS (<i>N</i> = 44) mean $\pm$ SD	Healthy children (<i>N</i> = 210) Mean $\pm$ SD	<i>t</i> value	<i>P</i> value	Effect size (<i>d</i>)
Physical functioning	70.17 $\pm$ 21.92	81.79 $\pm$ 15.34	− 3.35	0.002	2.53
Emotional functioning	69.32 $\pm$ 24.53	79.98 $\pm$ 16.92	− 2.75	0.008	0.43
Social functioning	76.70 $\pm$ 23.82	83.48 $\pm$ 22.83	− 1.78	0.077	0.28
Cognitive functioning	65.91 $\pm$ 23.90	77.80 $\pm$ 18.73	− 3.11	0.003	0.50
Communication	76.17 $\pm$ 20.77	86.53 $\pm$ 16.01	− 3.12	< 0.01	0.50
Worry	64.32 $\pm$ 24.72	78.03 $\pm$ 19.52	− 3.46	0.001	0.55
Parent HRQOL summary score	70.93 $\pm$ 19.99	80.76 $\pm$ 15.17	− 3.08	0.003	0.49
Daily activities	64.21 $\pm$ 30.58	73.31 $\pm$ 20.54	− 1.89	0.65	0.30
Family relationships	72.27 $\pm$ 24.08	79.69 $\pm$ 18.23	− 1.93	0.019	0.31
Family functioning summary score	68.24 $\pm$ 25.08	76.50 $\pm$ 17.35	− 2.08	0.042	0.33
Total score	70.26 $\pm$ 20.68	80.07 $\pm$ 13.98	− 3.01	0.004	0.47

SBS short bowel syndrome, *SD* standard deviation, *FIM* Family Impact Module

*The sample size of the control group for norm values of PedsQL 3.0 FIM is 210, and we only compared the means of children aged 5–12 years in our study with the children at the same age in the literature

Table 5 Comparing means of scale scores and total scores of PedsQL between different subgroups according to PHQ-9 or ZBI score

Variables	Major depression			Categorized burden score			
	Yes	No	<i>P</i> value	No or mild	Moderate	Severe	<i>P</i> value
1–24 months							
Physical health summary	54.21 ± 13.52	72.41 ± 14.12	0.010	71.15 ± 15.94	75.00 ± 16.45	59.66 ± 12.52	0.115
Emotional functioning	53.82 ± 21.18	73.13 ± 16.36	0.026	73.75 ± 15.47	75.30 ± 22.64	57.87 ± 16.59	0.104
Social functioning	77.50 ± 26.97	87.56 ± 13.99	0.414	89.25 ± 12.01	85.35 ± 20.02	80.69 ± 21.71	0.594
Cognitive functioning	74.08 ± 26.21	74.24 ± 24.50	0.989	72.43 ± 26.67	77.08 ± 22.21	73.92 ± 25.92	0.932
Psychosocial health summary	68.47 ± 19.11	78.31 ± 16.44	0.226	78.48 ± 16.32	79.25 ± 19.85	70.83 ± 16.86	0.314
Total score	62.76 ± 15.77	75.95 ± 15.01	0.074	75.55 ± 15.80	77.55 ± 18.39	66.36 ± 13.43	0.550
2–18 years							
Physical functioning	66.58 ± 19.24	85.42 ± 16.28	0.001	85.17 ± 17.18	86.56 ± 10.32	66.59 ± 19.86 ^{b,c}	0.003
Emotional functioning	66.92 ± 21.56	79.21 ± 15.43	0.019	80.11 ± 13.61	80.50 ± 17.17	62.69 ± 21.95 ^b	0.003
Social functioning	65.38 ± 24.36	79.04 ± 21.70	0.049	79.36 ± 23.44	78.50 ± 17.80	64.62 ± 20.56	0.110
School functioning [†]	55.00 ± 15.81	70.54 ± 18.09	0.028	72.59 ± 17.92	58.00 ± 13.04	53.13 ± 14.62 ^b	0.009
Psychosocial health summary	64.04 ± 19.64	76.39 ± 13.68	0.048	76.90 ± 12.77	76.92 ± 17.22	61.79 ± 18.55 ^{b,c}	0.005
Total score	64.37 ± 18.36	78.82 ± 12.50	0.001	79.12 ± 12.20	79.65 ± 13.74	62.68 ± 17.22 ^{b,c}	0.001

[†] 51 children had already been taken to school

^a Post hoc significance between no or mild and moderate

^b Post hoc significance between no or mild and severe

^c Post hoc significance between moderate and severe

functioning dimensions. In fact, the impact of SBS on school functionality has been previously reported in other studies [25]. Therefore, we suggest adopting the intermittent parenteral nutrition infusion strategy if possible. Specifically, increased stool frequency, loose stools, and abdominal pain were commonly observed in patients with SBS. Our research had revealed increased frequency of bowel movements seriously affects the QoL of the children. Similarly, one study also found that half of the parents of IF toddlers reported a decline in their children's HRQoL due to diarrhea [25]. Wong M et al. reported that HRQoL scores improved with age [22]. One interpretation is that as children with IF mature, they progressively adapt to and integrate the daily care routines necessary for effective IF management into their normal lives. In our study, the quality of life was not higher in the subgroup using teduglutide owing to the severe disease. Chen K et al. found that the impact of teduglutide treatment on SBS-related QoL was significant and most pronounced among patients with highest baseline parenteral support volume requirement [26]. Further research is needed to explore the long-term impact of teduglutide on QoL.

Families with children with SBS demonstrated significantly lower scores across nearly all dimensions of PedsQL FIM compared to healthy control families. The physical functioning dimension exhibited the most significant score difference (−11.6) compared with the reference population. This may be attributed to the overnight EN plus PN support, which could potentially impair the physical health dimension [27]. Although no significant differences were observed in PedsQL

GCM between children on PN and those weaned off PN, families of children on PN reported considerable negative impacts. For instance, these families exhibited significantly lower scores in the emotional functioning, social functioning, worry, and daily activities, as well as in the overall total score. The impact of PN infusion on families is considerable, as the level of care required for children on PN is substantial. Neam et al. found that PN dependence was independently associated with lower scores in the communication (−7, $p=0.03$) and daily activities (−10, $p=0.02$) domains [6]. Thomassen et al. [21] also reported similar results, indicating that PN dependency did not influence HRQoL scores. However, families with children dependent on PN exhibited a lower FIM score compared to those with children weaned from PN (49.9 vs. 73.6, $p<0.001$). In multivariate logistic regression analysis, younger patient age was independently associated with lower total scores of FIM ($p<0.05$). Consistent with our research, Neam et al. prospectively evaluated HFSUPPLRQoL in 117 families and demonstrated that the FIM total score increased with patient age [6]. The early stage of the disease is particularly challenging for families, as they must come to terms with the reality that their child has complex medical needs. As the child grows, families gradually integrate care giving routines into their daily lives to meet the needs of a child with SBS, these routines often involve organizing daily activities around dietary requirements, PN administration, and the avoidance of activities that may compromise the child's health. In addition, families may respond to their child's condition by improving communication and placing greater emphasis on the improving survival rates

in pediatric IF. Understanding how medical services and supports impact care would be valuable information that can help determine what supports and services could be developed in partnership with families [28]. In conclusion, it is essential to incorporate psychological support into intestinal rehabilitation teams. Teduglutide treatment associated reduction in parenteral support volume was accompanied by reduced infusion times, sleep disturbances, stable oral intake surrogates, and improved stool characteristics, all of these potential parameters for improving quality of life [29]. When comparing the use of teduglutide subgroup with the no use of teduglutide subgroup, there was no statistically significant difference in the mean total score of FIM. The reason is that the 8 patients had a longer duration of parenteral nutrition.

Burden and depression among caregivers

In our study, the families of children with SBS faced considerable economic pressure. Half of the families with one child diagnosed with SBS reported out-of-pocket expenses exceeding €24,400, while nearly half of these households earned less than €12,200 annually. Univariate analysis also revealed that caregivers who did not perceive a financial burden reported higher mean total scale of the PedsQL. Similarly, a Spanish study [30] indicated that the mean annual direct costs per child amounted to €74,734, with PN and hospitalizations identified as the primary cost drivers. Non-healthcare costs and indirect costs, resulting from productivity loss, also contribute to the overall economic burden [31]. Interventions that have the potential to enhance QoL for both patients with SBS and caregivers include patient education, participation in support groups, management of concomitant symptoms, and pharmacotherapies aimed at reducing the reliance on PN [31].

We found 40.6% of caregivers of children with SBS experienced a moderately to severe burden, as measured by the ZBI survey, and 19.8% met the criteria for clinical depression. A meta-analysis even reported an even higher prevalence, with up to 35% of parents of children with chronic illnesses meeting the criteria for clinical depression [32]. Higher scores on the PHQ-9 and the ZBI total scales were significantly negatively correlated with scores on the PedsQL GCM in our study. Neumann ML et al. found that sleep disturbances, insufficient support, and psychological stressors were reported more frequently as stressors than logistical challenges associated with caregiving, such as managing therapies and preparing specialized meals [33]. These findings underscore the importance of developing interventions aimed at improving the health and well-being of parents of children with SBS [32]. Furthermore, our research highlights the necessity of implementing targeted screening for depression and caregiver burden among parents of children with SBS.

Strengths and limitations

Our study has several strengths. It is the first study to evaluate the QoL of these pediatric patients diagnosed with SBS in China. The sample size was relatively large, comprising 96 children with SBS, which exceeds that of most previous studies. Additionally, a comprehensive range of factors associated with QoL, including disease-related characteristics and clinical parameters, was evaluated. Considering the differences between China and developed countries in terms of social, economic, and medical characteristics, it is both necessary to assess the QoL of children with SBS and their caregivers in China.

This study has several limitations. First, as a single-center study, our findings may lack generalizability, and the results from multicenter studies would likely offer greater external validity. Second, assessing HRQoL in children presents a well-documented challenge. The difference in HRQoL between patient and proxy reports is a well-known phenomenon [34, 35]. Third, the use of self-report questionnaires limits its diagnostic certainty, as PHQ scores indicate depressive symptomatology rather than confirmed diagnosis. Therefore, future studies should incorporate structured clinical interviews to confirm depressive disorders and strengthen the clinical validity of the findings.

Conclusions

This study provides a comprehensive analysis of proxy-reported HRQoL among children with SBS in China and identifies some independent predictors (perceived no financial burden and short-term of PN). The independent predictors of a higher total score on the PedsQL FIM were older age, female gender, short duration of hospitalization and perceived no financial burden. We highlighted that the HRQoL of children with SBS and their caregivers was significantly lower compared to that of healthy controls published in previous studies. The caregiver's high burden and depression showed a negative correlation with the children's HRQoL. Therefore, integrating psychological support into intestinal rehabilitation teams is essential. Addressing the needs of family caregivers should be considered an integral component of comprehensive care for pediatric patients with SBS.

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Authors' contributions Guannan Bai and Ying Wang designed the study. Haixia Feng, Qingqing Wu, Shanshan Chen, Qinzhou Xu, Shiyi Liu, Wei Cai and Ying Wang enrolled patients and collected data. Haixia Feng, Qingqing Wu, Chen Cai and Weihui Yan analyzed the data. Haixia Feng and Qingqing Wu wrote the paper. Haixia Feng, Qingqing Wu, Chen Cai, Weihui Yan, Shanshan Chen, Qinzhou Xu, Shiyi Liu, Wei Cai, Guannan Bai and Ying Wang contributed to

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Data availability All data supporting the findings of this study are available within the paper and its Supplementary Information.

Declarations

Ethical approval and consent to participate The study was approved by the Ethics Committee of Xinhua Hospital Affiliated to Shanghai Jiao Tong University School of Medicine (XHEC-C-2020-073) and conformed to the ethics guidelines of the Declaration of Helsinki. Written informed consent was obtained from all participants in the study.

Competing interests The authors declare no competing interests.

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Authors and Affiliations

Haixia Feng¹ · Qingqing Wu¹ · Chen Cai¹ · Weihui Yan^{1,3} · Shanshan Chen¹ · Qinzhou Xu¹ · Shiyi Liu¹ · Wei Cai^{2,3} · Guannan Bai⁴ · Ying Wang^{1,3}

✉ Wei Cai
caiw204@sjtu.edu.cn

✉ Guannan Bai
guannanbai@zju.edu.cn

✉ Ying Wang
wangying02@xinhumed.com.cn

¹ Division of Pediatric Gastroenterology and Nutrition, School of Medicine, Xinhua Hospital, Shanghai Jiao Tong University, Shanghai 200092, China

² Shanghai Institute for Pediatric Research, Shanghai 200092, China

³ Shanghai Key Laboratory of Pediatric Gastroenterology and Nutrition, Shanghai 200092, China

⁴ Department of Child Health Care, Children's Hospital, Zhejiang University School of Medicine, National Clinical Research Center for Child Health, Hangzhou 310052, China